

# librispeech\_validated\_voters\_ensemble

## scalars\_librispeech\_validated\_voters\_ensemble

| key                                                        | value                                      |
|------------------------------------------------------------|--------------------------------------------|
| n_samples                                                  | 2611                                       |
| main_kwargs_applied                                        | {}                                         |
| timings_seconds.cross_wer                                  | 0.7                                        |
| timings_seconds.ranking                                    | 23.2                                       |
| timings_seconds.model_filtration                           | 5.1                                        |
| timings_seconds.rank_distribution                          | 27                                         |
| ranking_stats.n_models                                     | 5                                          |
| ranking_stats.pool                                         | seamless_m4t+whisper_v3+parakeet+sb+cohere |
| rank_distribution_summary.n_subpools                       | 6                                          |
| rank_distribution_summary.n_boot_inner                     | 1000                                       |
| rank_distribution_summary.n_fusions                        | 16                                         |
| rank_distribution_summary.n_scorings                       | 25                                         |
| rank_distribution_summary.seconds_fuse_and_score           | 21.7                                       |
| rank_distribution_summary.seconds_bootstrap                | 0.3                                        |
| rank_distribution_summary.weighting                        | filtration                                 |
| rank_distribution_summary.exponent                         | 1                                          |
| rank_distribution_summary.mean_entropy                     | 0.3378                                     |
| rank_distribution_summary.max_entropy                      | 0.4755                                     |
| rank_distribution_summary.n_models_modal_differs_from_full | 0                                          |

## ranking\_librispeech\_validated\_voters\_ensemble

| model        | role  | n_samples | wer_pseudo_loo | cer_pseudo_loo | rank_pseudo |
|--------------|-------|-----------|----------------|----------------|-------------|
| cohere       | voter | 2611      | 1.0833         | 0.3562         | 1           |
| parakeet     | voter | 2611      | 1.1573         | 0.3496         | 2           |
| sb           | voter | 2611      | 1.6315         | 0.4992         | 3           |
| seamless_m4t | voter | 2611      | 1.9065         | 0.7159         | 4           |
| whisper_v3   | voter | 2611      | 2.0014         | 1.1836         | 5           |

## model\_confidence\_librispeech\_validated\_voters\_ensemble

| model    | rank_pseudo_full | rank_modal | p_modal | p_at_full_rank | entropy_norm | rank_mean | rank_sd | rank_ci_lo | rank_ci_hi | agrees_with_full |
|----------|------------------|------------|---------|----------------|--------------|-----------|---------|------------|------------|------------------|
| cohere   | 1                | 1          | 0.7621  | 0.7621         | 0.3409       | 1.2379    | 0.4258  | 1          | 2          | True             |
| parakeet | 2                | 2          | 0.7621  | 0.7621         | 0.3409       | 1.7621    | 0.4258  | 1          | 2          | True             |

| model        | rank_pseudo_full | rank_modal | p_modal | p_at_full_rank | entropy_norm | rank_mean | rank_sd | rank_ci_lo | rank_ci_hi | agrees_with_full |
|--------------|------------------|------------|---------|----------------|--------------|-----------|---------|------------|------------|------------------|
| sb           | 3                | 3          | 0.9547  | 0.9547         | 0.1179       | 3.0465    | 0.2157  | 3          | 4          | True             |
| seamless_m4t | 4                | 4          | 0.6576  | 0.6576         | 0.414        | 4.3335    | 0.4808  | 4          | 5          | True             |
| whisper_v3   | 5                | 5          | 0.6609  | 0.6609         | 0.4755       | 4.62      | 0.5634  | 3          | 5          | True             |

### rank\_distribution\_librispeech\_validated\_voters\_ensemble

| model        | rank_1 | rank_2 | rank_3 | rank_4 | rank_5 |
|--------------|--------|--------|--------|--------|--------|
| seamless_m4t | 0      | 0      | 0.0044 | 0.6576 | 0.3379 |
| whisper_v3   | 0      | 0      | 0.0409 | 0.2981 | 0.6609 |
| parakeet     | 0.2379 | 0.7621 | 0      | 0      | 0      |
| sb           | 0      | 0      | 0.9547 | 0.0442 | 0.0011 |
| cohere       | 0.7621 | 0.2379 | 0      | 0      | 0      |

### pair\_probability\_librispeech\_validated\_voters\_ensemble

| pair                    | pred_winner_full | gap_wer_pseudo | p_first_beats_second | p_supports_full |
|-------------------------|------------------|----------------|----------------------|-----------------|
| seamless_m4t whisper_v3 | seamless_m4t     | 0.0949         | 0.6608               | 0.6608          |
| parakeet cohere         | cohere           | 0.0741         | 0.2379               | 0.7621          |
| whisper_v3 sb           | sb               | 0.3699         | 0.0409               | 0.9591          |
| seamless_m4t sb         | sb               | 0.275          | 0.0055               | 0.9945          |
| seamless_m4t parakeet   | parakeet         | 0.7492         | 0                    | 1               |
| seamless_m4t cohere     | cohere           | 0.8232         | 0                    | 1               |
| whisper_v3 parakeet     | parakeet         | 0.8441         | 0                    | 1               |
| whisper_v3 cohere       | cohere           | 0.9182         | 0                    | 1               |
| parakeet sb             | parakeet         | 0.4741         | 1                    | 1               |
| sb cohere               | cohere           | 0.5482         | 0                    | 1               |

### model\_filtration\_librispeech\_validated\_voters\_ensemble

| model        | n_subpools | f_bar  | f_sd   | f_min  | f_max  | survival |
|--------------|------------|--------|--------|--------|--------|----------|
| cohere       | 5          | 0.0512 | 0.0089 | 0.0448 | 0.0655 | 0.9488   |
| parakeet     | 5          | 0.0631 | 0.0069 | 0.0563 | 0.0743 | 0.9369   |
| whisper_v3   | 5          | 0.0972 | 0.0109 | 0.0885 | 0.1099 | 0.9028   |
| sb           | 5          | 0.1148 | 0.0111 | 0.1038 | 0.1325 | 0.8852   |
| seamless_m4t | 5          | 0.1619 | 0.0154 | 0.144  | 0.1838 | 0.8381   |

### by\_pool\_size\_librispeech\_validated\_voters\_ensemble

| k | pools | total_weight | mean_entropy | max_entropy |
|---|-------|--------------|--------------|-------------|
| 4 | 5     | 0.8          | 0.261        | 0.377       |
| 5 | 1     | 0.2          | 0.2548       | 0.2548      |

### cross\_wer\_librispeech\_validated\_voters\_ensemble

|              | seamless_m4t | whisper_v3 | parakeet | sb   | cohere |
|--------------|--------------|------------|----------|------|--------|
| seamless_m4t | 0            | 2.84       | 2.14     | 2.56 | 2.14   |
| whisper_v3   | 2.84         | 0          | 2.22     | 2.81 | 2.16   |
| parakeet     | 2.14         | 2.22       | 0        | 1.88 | 1.29   |
| sb           | 2.56         | 2.81       | 1.88     | 0    | 1.74   |
| cohere       | 2.14         | 2.16       | 1.29     | 1.74 | 0      |

### subpool\_filtration\_librispeech\_validated\_voters\_ensemble

| pool                                       | k | model        | n_samples | filtered_out | filtered_out_rate |
|--------------------------------------------|---|--------------|-----------|--------------|-------------------|
| seamless_m4t+whisper_v3+parakeet+sb        | 4 | seamless_m4t | 2611      | 376          | 0.144             |
| seamless_m4t+whisper_v3+parakeet+sb        | 4 | whisper_v3   | 2611      | 231          | 0.0885            |
| seamless_m4t+whisper_v3+parakeet+sb        | 4 | parakeet     | 2611      | 147          | 0.0563            |
| seamless_m4t+whisper_v3+parakeet+sb        | 4 | sb           | 2611      | 293          | 0.1122            |
| seamless_m4t+whisper_v3+parakeet+cohere    | 4 | seamless_m4t | 2611      | 428          | 0.1639            |
| seamless_m4t+whisper_v3+parakeet+cohere    | 4 | whisper_v3   | 2611      | 234          | 0.0896            |
| seamless_m4t+whisper_v3+parakeet+cohere    | 4 | parakeet     | 2611      | 159          | 0.0609            |
| seamless_m4t+whisper_v3+parakeet+cohere    | 4 | cohere       | 2611      | 142          | 0.0544            |
| seamless_m4t+whisper_v3+sb+cohere          | 4 | seamless_m4t | 2611      | 394          | 0.1509            |
| seamless_m4t+whisper_v3+sb+cohere          | 4 | whisper_v3   | 2611      | 234          | 0.0896            |
| seamless_m4t+whisper_v3+sb+cohere          | 4 | sb           | 2611      | 282          | 0.108             |
| seamless_m4t+whisper_v3+sb+cohere          | 4 | cohere       | 2611      | 118          | 0.0452            |
| seamless_m4t+parakeet+sb+cohere            | 4 | seamless_m4t | 2611      | 436          | 0.167             |
| seamless_m4t+parakeet+sb+cohere            | 4 | parakeet     | 2611      | 156          | 0.0597            |
| seamless_m4t+parakeet+sb+cohere            | 4 | sb           | 2611      | 271          | 0.1038            |
| seamless_m4t+parakeet+sb+cohere            | 4 | cohere       | 2611      | 121          | 0.0463            |
| whisper_v3+parakeet+sb+cohere              | 4 | whisper_v3   | 2611      | 287          | 0.1099            |
| whisper_v3+parakeet+sb+cohere              | 4 | parakeet     | 2611      | 168          | 0.0643            |
| whisper_v3+parakeet+sb+cohere              | 4 | sb           | 2611      | 307          | 0.1176            |
| whisper_v3+parakeet+sb+cohere              | 4 | cohere       | 2611      | 117          | 0.0448            |
| seamless_m4t+whisper_v3+parakeet+sb+cohere | 5 | seamless_m4t | 2611      | 480          | 0.1838            |
| seamless_m4t+whisper_v3+parakeet+sb+cohere | 5 | whisper_v3   | 2611      | 283          | 0.1084            |
| seamless_m4t+whisper_v3+parakeet+sb+cohere | 5 | parakeet     | 2611      | 194          | 0.0743            |
| seamless_m4t+whisper_v3+parakeet+sb+cohere | 5 | sb           | 2611      | 346          | 0.1325            |

| pool                                       | k | model  | n_samples | filtered_out | filtered_out_rate |
|--------------------------------------------|---|--------|-----------|--------------|-------------------|
| seamless_m4t+whisper_v3+parakeet+sb+cohere | 5 | cohere | 2611      | 171          | 0.0655            |

### subpool\_ranks\_librispeech\_validated\_voters\_ensemble (first 60 of 67 rows)

| pool                                    | k | model        | rank | count |
|-----------------------------------------|---|--------------|------|-------|
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | seamless_m4t | 3    | 25    |
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | seamless_m4t | 4    | 790   |
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | seamless_m4t | 5    | 185   |
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | whisper_v3   | 3    | 64    |
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | whisper_v3   | 4    | 128   |
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | whisper_v3   | 5    | 808   |
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | parakeet     | 2    | 1000  |
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | sb           | 3    | 911   |
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | sb           | 4    | 82    |
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | sb           | 5    | 7     |
| seamless_m4t+whisper_v3+parakeet+sb     | 4 | cohere       | 1    | 1000  |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | seamless_m4t | 4    | 570   |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | seamless_m4t | 5    | 430   |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | whisper_v3   | 3    | 48    |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | whisper_v3   | 4    | 382   |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | whisper_v3   | 5    | 570   |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | parakeet     | 1    | 376   |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | parakeet     | 2    | 624   |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | sb           | 3    | 952   |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | sb           | 4    | 48    |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | cohere       | 1    | 624   |
| seamless_m4t+whisper_v3+parakeet+cohere | 4 | cohere       | 2    | 376   |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | seamless_m4t | 3    | 3     |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | seamless_m4t | 4    | 712   |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | seamless_m4t | 5    | 285   |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | whisper_v3   | 3    | 46    |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | whisper_v3   | 4    | 239   |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | whisper_v3   | 5    | 715   |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | parakeet     | 1    | 918   |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | parakeet     | 2    | 82    |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | sb           | 3    | 951   |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | sb           | 4    | 49    |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | cohere       | 1    | 82    |
| seamless_m4t+whisper_v3+sb+cohere       | 4 | cohere       | 2    | 918   |

| pool                                       | k | model        | rank | count |
|--------------------------------------------|---|--------------|------|-------|
| seamless_m4t+parakeet+sb+cohere            | 4 | seamless_m4t | 4    | 520   |
| seamless_m4t+parakeet+sb+cohere            | 4 | seamless_m4t | 5    | 480   |
| seamless_m4t+parakeet+sb+cohere            | 4 | whisper_v3   | 3    | 33    |
| seamless_m4t+parakeet+sb+cohere            | 4 | whisper_v3   | 4    | 447   |
| seamless_m4t+parakeet+sb+cohere            | 4 | whisper_v3   | 5    | 520   |
| seamless_m4t+parakeet+sb+cohere            | 4 | parakeet     | 1    | 107   |
| seamless_m4t+parakeet+sb+cohere            | 4 | parakeet     | 2    | 893   |
| seamless_m4t+parakeet+sb+cohere            | 4 | sb           | 3    | 967   |
| seamless_m4t+parakeet+sb+cohere            | 4 | sb           | 4    | 33    |
| seamless_m4t+parakeet+sb+cohere            | 4 | cohere       | 1    | 893   |
| seamless_m4t+parakeet+sb+cohere            | 4 | cohere       | 2    | 107   |
| whisper_v3+parakeet+sb+cohere              | 4 | seamless_m4t | 4    | 716   |
| whisper_v3+parakeet+sb+cohere              | 4 | seamless_m4t | 5    | 284   |
| whisper_v3+parakeet+sb+cohere              | 4 | whisper_v3   | 3    | 18    |
| whisper_v3+parakeet+sb+cohere              | 4 | whisper_v3   | 4    | 266   |
| whisper_v3+parakeet+sb+cohere              | 4 | whisper_v3   | 5    | 716   |
| whisper_v3+parakeet+sb+cohere              | 4 | parakeet     | 1    | 15    |
| whisper_v3+parakeet+sb+cohere              | 4 | parakeet     | 2    | 985   |
| whisper_v3+parakeet+sb+cohere              | 4 | sb           | 3    | 982   |
| whisper_v3+parakeet+sb+cohere              | 4 | sb           | 4    | 18    |
| whisper_v3+parakeet+sb+cohere              | 4 | cohere       | 1    | 985   |
| whisper_v3+parakeet+sb+cohere              | 4 | cohere       | 2    | 15    |
| seamless_m4t+whisper_v3+parakeet+sb+cohere | 5 | seamless_m4t | 4    | 643   |
| seamless_m4t+whisper_v3+parakeet+sb+cohere | 5 | seamless_m4t | 5    | 357   |
| seamless_m4t+whisper_v3+parakeet+sb+cohere | 5 | whisper_v3   | 3    | 38    |
| seamless_m4t+whisper_v3+parakeet+sb+cohere | 5 | whisper_v3   | 4    | 319   |